

Sri Siddharoodha Swamy Temple Digital Management System

Simplified Project Reference & Structure Guide

1. What is this Project?

This project is a website and backend server designed for the Sri Siddharoodha Swamy Temple in Hubballi, Karnataka. It replaces old paper registers with a digital interface. It helps devotees register, book rituals (Sevas), reserve rooms for staying, make secure donations, and watch live temple feeds.

By doing this online, physical queues are reduced from 30 minutes to under 2 minutes, and errors like booking the same room twice are eliminated.

2. How the Project is Structured (3-Tier)

The project is split into three parts that work together:

- **The Frontend (User Interface):** The website that devotees see in their web browsers. Built with React.
- **The Backend (Server Logic):** The engine that runs behind the scenes. It handles calculations, checks security, and sends emails. Built with Spring Boot.
- **The Database (Storage):** The safe filing cabinet that stores user passwords, room status, bookings, and donation logs. Built with MongoDB.

3. Frontend Packages & Technologies

The frontend runs in the web browser using these main libraries:

- **React:** Helps build the visual pages (Home, Sevas, Prasad, Rooms) using components.
- **Vite:** The tool that compiles and runs the frontend development server very fast.
- **Tailwind CSS:** Used for colors, layout, and mobile-friendly screens.
- **Axios:** Sends the user's booking data securely to the backend server.
- **Framer Motion:** Creates smooth sliding and fading animations for a premium feel.
- **Lucide React:** Provides clean icons for buttons and navigations.

4. Backend Packages & Technologies

The backend is written in Java 21 and Spring Boot, using these main libraries:

- **Spring Boot Starter Web:** Enables the backend to receive HTTP requests and reply in JSON.

- **Spring Data MongoDB:** Connects Java to the MongoDB database automatically.
- **Spring Security & JWT (JSON Web Tokens):** Secures critical API pages. Users log in, get an encrypted token, and use it to access endpoints securely.
- **Spring Boot Mail:** Auto-generates and emails beautifully styled HTML transaction receipts directly to the devotee.
- **Lombok:** Auto-generates common code behind the scenes to keep files neat.

5. Database (MongoDB)

Instead of relational tables (like in MySQL), the database uses flexible collections (storing JSON-like documents). The main tables (collections) used are:

- **users:** Devotee email, encrypted password, phone, and role permissions.
- **bookings:** Booked Sevas lists, Gotra/Nakshatra, performance date, and price details.
- **room_bookings:** Guest list count, room code, check-in, check-out dates, and payment status.
- **donations:** Contributor name, category (like food or cow shelter), amount, and transaction code.